Predicting Chanterelle Observations with Place Names

Marcus Bennevall
Stockholm University

Abstract

This study investigates whether linguistic features in Swedish place names can improve predictions of chanterelle observations beyond conventional environmental variables. Comparing logistic regression models with and without such linguistic features, I find that the linguistic models consistently achieve higher Average Precision (AP) than the best control model across temporal, spatial, and spatiotemporal evaluations. A sparse TF-IDF representation of place names boosts AP by 20.5-72.6%, while improvements by a dense fastText representation are limited to 0.5-3.9%. TF-IDF coefficient analysis suggests that the predictive value of place names is derived from both local landscape features and larger language patterns.

1 Introduction

Cantharellus cibarius, the golden chanterelle, is an edible mushroom widely appreciated in Sweden for its taste and appearance. Machine learning methods have thus been applied in attempts to predict where they may sprout and thrive, for example by Frank-Hansen (2024) and Kantarellappen (2026). These approaches typically make predictions based on environmental features – forest coverage, temperature, precipitation, etc. – with moderate success.

However, researchers in other areas have found linguistic information in place names to be a surprisingly useful predictor of various ecological patterns, including the historic ranges of wolves and bears in Italy (Tattoni, 2019), locations of oyster populations off the Scottish coast (McGregor et al., 2025), and historic forest fires in northern Sweden (Cogos et al., 2019). The current study aims to extend this line of research by investigating whether similar success can be achieved in the context of Swedish chanterelles.

Specifically, my research question is this: **Can linguistic features in Swedish place names improve predictions of chanterelle observations beyond conventional environmental features?** My hypothesis, given the research above, is yes. In what follows, I will describe how I used six logistic regression models to explore the predictive value of place names in the context of chanterelle observations, evaluating the predictions temporally, spatially, and spatiotemporally using average precision.[1] The results include three central findings:

1. Place name information does improve predictions beyond environmental features.
2. The linguistic content of place names matters beyond their density, with a sparse TF-IDF representation performing substantially better than a dense representation based on fastText embeddings.
3. Predictions benefit from both local landscape features and regional language patterns in place names.

From the start, it is important to emphasize that my models predict *observations* of chanterelles, not actual chanterelles. While predictions of live chanterelles would have been more useful, such datasets are unfortunately not available at scale. I use observations as a workable but admittedly flawed proxy: Compared to real chanterelles, chanterelle observations are more likely in human-accessible areas and less likely in remote and inhospitable places.

2 Method

Conceptually, the project can be divided into two roughly equal parts – data processing and model experiments – where a shared challenge was computer memory. For example, the raw data and pretrained fastText embedding model alone required around 20 GBs of disk space, while the large feature sets of the TF-IDF and fastText prediction

[1] The full code is available on GitHub (Hazelime, 2026).

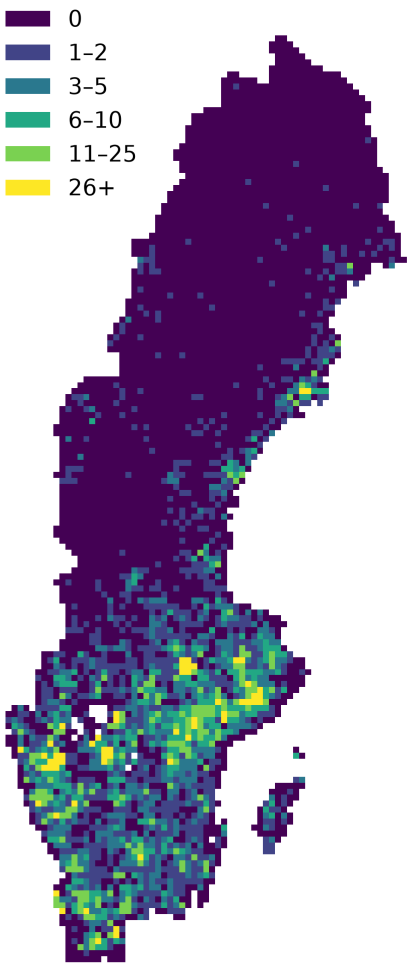


Figure 1: Number of positive cell-weeks 1995-2025.

models necessitated a custom-built logistic regression model due to RAM constraints. I therefore spent much time learning about and working on file compression and computational efficiency.

2.1 Data

All sources of input data described below were eventually linked to a 10 x 10 km master grid of Sweden built from the joint NMD2023 and NDM2018 10 x 10 m raster (v0.3) in the National Land Cover Database (Naturvårdsverket, 2026). For the observations and weather data, this required a transformation from a latitude- and longitude coordinate reference system (CRS) to the meter-based one used by the raster (EPSG:3006). After excluding water-only cells (raster codes 62 and 63), the grid consisted of 4,983 cells. The **forest coverage** data was simultaneously calculated per cell as the percentage of land tiles in the raster coded as forest or alpine trees (raster codes 23, 43, 110, and 120).

The **chanterelle observations** data was downloaded via API from the SLU Swedish Species Information Centre (2026). A total of 15,964 recorded observations of *Cantharellus cibarius* were reduced to 12,943 after filtering for observations within my defined grid, location uncertainty ≤ 5 km (so they could be reasonably assigned to a grid cell), and the years 1995-2025. I excluded 2026 because that data was not yet complete, and I chose 1995 as my starting cut-off as that was when the yearly data started to include hundreds of observations consistently. Assigning each observation to a grid cell and an ISO week, I ended up with a total of 10,312 positive cell-weeks among 8,057,511 cell-weeks total. The positive prevalence, i.e. fraction of positives, was thus very low: 0.128%. Figure 1 shows the geographical distribution.

The **precipitation** and **temperature** data was downloaded manually year by year from SMHI (2025). Since I used ISO weeks, which run Monday-Sunday, I additionally retrieved 1994 data to ensure complete coverage of the first week of 1995. The weather data was recorded by 4 x 4 km cell, so to align it with the master grid I first assigned each weather cell to a grid cell by its centroid and then averaged the data points within each grid cell. Two grid cells that lacked weather data were assigned the same data as their nearest neighbor; otherwise, the data offered complete coverage across all weeks and cells. From this daily cell data, I calculated weekly total precipitation and weekly average temperature.

The **place name** data was downloaded manually from Lantmäteriet (2025) upon being granted authorization, which was near-immediate. After 246 off-grid records were removed, 989,567 records were aggregated by grid cell ($\mu = 198.6, \sigma = 196.0$, max $= 3,960$). **Place name count** was then calculated per cell along with a log-transformed version $x_{log} = \ln(1 + x)$ meant to address right-skewness and the influence of very large counts.

The **week** data was encoded cyclically as the sine and cosine of the angle represented by the ISO week number in its annual cycle. This allowed for a more realistic representation where the first week of a year is conceptually close to the last week of the preceding year rather than far apart.

To reduce memory load, processed data was saved using efficient NumPy datatypes in zstd-compressed Parquet files.

2.2 Data splits

To enable validation and testing, data was split into separate training, validation, and test sets for

temporal, spatial, and spatiotemporal evaluation (Table 1). These schemes evaluated how well the models generalized to unseen years, unseen areas, and unseen areas during unseen years, respectively.

	Train	Validation	Test
Temporal	All blocks 1995-2023	All blocks 2024	All blocks 2025
Spatial	60 blocks All years	8 blocks All years	8 blocks All years
Spatio-temporal	Train blocks 1995-2023	Val blocks 2024	Test blocks 2025

Table 1: Data assignment by split and evaluation scheme. Spatiotemporal splits are intersections of corresponding temporal and spatial splits.

Since the spatial and spatiotemporal splits relied on randomization, I decided to group the 10 x 10 km cells into 100 x 100 km blocks to minimize the risk of data leakage between neighboring cells in different splits. For these evaluation schemes, I also averaged the results across five runs with different seeds (111, 222, 333, 444, and 555). However, it should be noted that the five seeds are not statistically independent, and moreover that not all blocks are alike, as some contain more land, observations, etc., than others.

2.3 Models

Six logistic regression models were compared to the **Positive prevalence**, which was a natural hard baseline since it represents the expected average precision of a random classifier. The first logistic-regression model, **Seasonality**, provided a soft baseline by including only the seasonal features `week_sin` and `week_cos`. The **Environmental** model then represented the soft baseline referenced in the research question by also incorporating the conventional environmental features `forest_pct` (forest coverage), `temp_7d`, and `precip_7d` (average temperature and total precipitation during the preceding week).

Placename_count and **Placename_count_log** functioned as control models since they both add a feature that measured place name density, either in absolute terms (`placename_count`) or logged (`placename_count_log`). Place name density here acted as a possible proxy of human activity, which was an important factor to control for since reason dictates that observations are more likely to be recorded where there are humans to record them.

The last two models constituted the main experiment, as they both expanded upon Placename_count_log's feature set with features that captured the linguistic information of a cell's place names. However, they did so in different ways. For **Tfidf**, each cell was represented as a document containing the cell's unique place names, from which 5,000 character n-gram TF-IDF features of length 3-5 were generated using Scikit-learn's TfidfVectorizer (Scikit-learn, 2026b). The vectorizer used the char_wb analyzer and was trained only on the respective training set in each evaluation scheme. I decided to deduplicate place names within cells because individual names often repeated for different place types (e.g., designating a municipality, a parish, a terrain, and a developed area all at once), and place name density was already accounted for.

For **Fasttext**, I downloaded a binary of a pretrained Swedish fastText model prepared by the original fastText authors (Facebook, 2018; Grave et al., 2018). I then generated 300-dimensional embeddings for all place names and, for each cell, averaged the embeddings of the unique place names therein. All linguistic cell-level features were L2-normalized. Using both TF-IDF and fastText allowed me to test how a sparse representation of place names would compare to a dense representation.

As far as possible, the logistic regression models were built identically. The baseline and control models all utilized Scikit-learn's LogisticRegression class with feature standardization, the recommended Limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS) solver, a convergence criterion of 1,000 max iterations, and default settings otherwise (Scikit-learn, 2026a). For the experimental models, however, this solution was prohibitively expensive. With a dataset of 8,057,511 cell-weeks multiplied by feature sets of size 306-5,006, an ensuing dense matrix would span billions of values and therefore require tens or even hundreds of GBs of RAM.

To address this issue, I built a custom logistic regression class which exploited the fact that the linguistic features were constant within each grid cell. Rather than repeating these features across all cell-weeks, this class modeled the predictor as

$$p_i = \sigma(b + x_i^\top \beta + v_{c(i)}^\top \gamma) \tag{1}$$

where σ is the sigmoid function, p_i is the probability of at least one chanterelle being observed during cell-week i, x_i contains the six numerical

features for that cell-week, $v_{c(i)}$ contains the 300-5,000 linguistic features for its associated grid cell, and β and γ are the respective learned coefficients. During training, the estimator could then minimize L2-regularized logistic loss while computing the linguistic contribution once per cell (4,983 times) instead of once per cell-week (8,057,511 times), drastically reducing the need for RAM.

2.4 Evaluation metric

The main evaluation metric I used was average precision (AP), which sorts predictions by probability and then averages precision@k for all ranks where there are actual observations. It thus quantifies the quality of the ranking. A simpler metric like accuracy would be misleading, since the extremely imbalanced dataset would make a majority-class baseline virtually unbeatable. Because AP calculates precision across all possible classification thresholds, it was also preferred over F-score, where you have to select a single threshold more or less arbitrarily.

2.5 Validation

The validation splits had three main purposes. First, they were used to select the L2 regularization parameter $C \in [0.01, 0.1, 1.0]$ that maximized validation AP for each model. Second, they helped determine which of the control models performed better. Tests showed that the model with a log-transformed place name count was superior across all evaluation schemes, so its feature set subsequently formed the foundation for the two experimental models. Third, the temporal validation set was used to set the convergence criteria of the custom logistic regression model. After testing function tolerance $\in [10^{-10}, 10^{-8}, 10^{-5}]$ and max iterations $\in [300, 1000, 1500]$, I finally settled on 10^{-8} and 1,000 since stricter convergence criteria produced identical results while laxer criteria yielded noticeable drops in temporal AP.

To ensure fair comparisons between Scikit-learn's LogisticRegression and my custom implementation, I compared the two implementations with 20 TF-IDF features on a 30 cells x 3000 weeks synthetic dataset. Since the maximum difference in prediction probabilities was $\leq 10^{-4}$, I considered the implementations numerically equivalent.

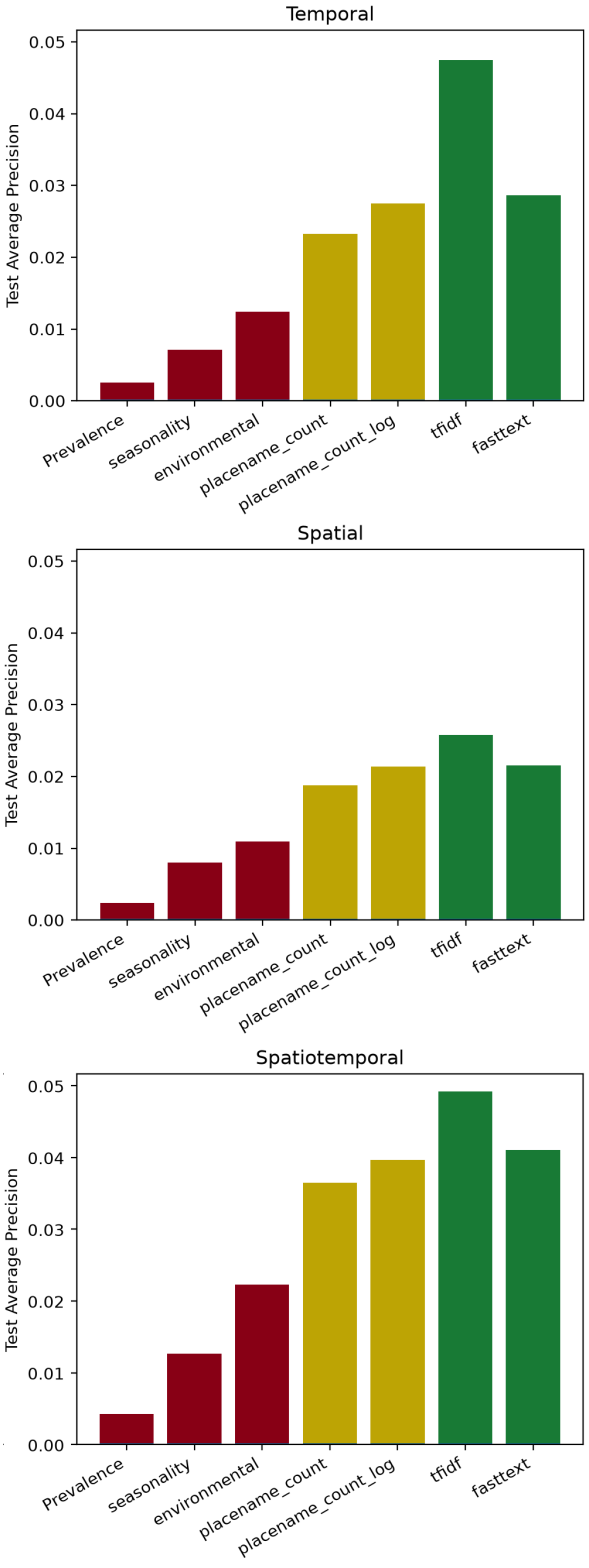


Figure 2: Test AP by model and evaluation scheme compared to positive prevalence. Baselines are in red, control models in yellow, and experimental models in green. The spatial and spatiotemporal diagrams show test AP averaged across five seeds.

3 Results and discussion

Figure 2 shows the results on the test splits across the three evaluation schemes. Test AP is overall very low in absolute terms (< 0.05), but this is to be expected with such a lopsided dataset and

thus difficult classification task. What is of interest is rather relative performance among models. For example, all logistic regression models outperform the positive prevalence baseline, and **both experimental models are superior to the environmental baseline**.

Particularly interesting for my research question is also that **both experimental models beat the best control model** (Table 2). This means that it is indeed the linguistic content of the place names that contribute to improved predictions rather than just place name density.

Model	Temporal	Spatial	Spatio-temporal
Placename_log_count	2.752	2.139	3.961
Tfidf	**4.748** (+72.5%)	**2.578** (+20.5%)	**4.916** (+24.1%)
Fasttext	2.860 (+3.9%)	2.150 (+0.5%)	4.107 (+3.7%)

Table 2: Test AP x 100 by evaluation scheme for selected models. AP values have been multiplied by 100 for easier comparison, and percentages refer to relative improvement compared to Placename_log_count. The spatial and spatiotemporal values are averages across five seeds.

However, it should be noted that **Tfidf shows far greater improvement than Fasttext** – especially in the temporal evaluation (+72.5% vs. +3.9%). In the spatial evaluation, Fasttext in fact achieves nearly identical test AP to Placename_count_log when averaged across five seeds (+0.5%).

One explanation for Tfidf's dominance over Fasttext may lie in how the aggregate per-cell features are constructed. With Fasttext, where each cell's features are produced by averaging the embedding features of all of its associated place names, it seems reasonable that linguistic information could get diluted in the process – especially with a cell resolution as coarse as 10 x 10 km. With Tfidf, in contrast, cell-level features are calculated without merging aspects of the individual place names.

Lastly, the fact that the two control models outclass the Environmental baseline model demonstrates the importance of a density feature in this context and supports the validity of the control models' inclusion. However, had I predicted actual chanterelles rather than chanterelle observations, it is unclear what role place name density would have played (if any). Exploring this feature thus constitutes a promising avenue for future research.

3.1 Linguistic Patterns in TF-IDF Coefficients

To better understand why TF-IDF outperforms Fasttext, I extracted the n-grams for the top 20 TF-IDF coefficients positively and negatively associated with chanterelle observations in the temporal evaluation scheme (which covers all cells).[2] Some recognizable positive patterns among these include *kulle* (hill), *mad* (wet meadow near a beach), *mölla* (mill), and *park* (park), whereas the negative coefficients are dominated by n-gram patterns such as *myr* (marsh), *jávri* (Sámi for lake), *moss* (peatland), and *holm* (islet).

Given that chanterelles prefer shaded, moderately wet, and sandy soils (Kumar et al., 2024), the environmental patterns above should come as no surprise even though the model predicts observations rather than actual chanterelles. What is more interesting are the regional linguistic patterns encoded in the words: *mad* and *mölla* are used predominantly in southern Sweden (the rest of the country would say *strandäng* and *kvarn*), whereas *jávri* – a Sámi word – is used exclusively in the north. Since there is a clear north-south divide in the observation data (see Figure 1), it seems reasonable to expect that a prediction strategy indirectly built around latitude estimation could prove very useful.

As a particularly striking example, consider the positive pattern *mölla* (mill in southern Sweden). Unlike the other top patterns, a mill is not a natural feature, but a building. While especially watermills may function well in landscapes that chanterelles also enjoy (and thus act as a kind of environmental proxy), I think the more important lesson here relates to the north-south distinction. Indeed, this distinction seems to hold so much predictive value that a dialectal word not obviously connected to nature, such as *mölla*, is able to outrank many patterns more clearly linked to mushroom ecology.

3.2 Influence of randomization

Since both the spatial and spatiotemporal evaluation relied on randomization, there was a certain amount of variation across seeds. For the spatial evaluation, the test AP standard deviation ranged from 0.0012 (Seasonality) to 0.0056 (Placename_count_log), whereas for the spatiotemporal evaluation it ranged from 0.0051 (Seasonality) to 0.0334

[2] Please see Appendix A for the full list.

(Tfidf). Please see Appendix B for more details.

One reason for the fairly large standard deviation in the spatiotemporal scheme is the comparatively low number of positive cell-weeks assigned to individual seeds there. This is a result of observations being constrained both to a certain year (2025) and to certain 100 x 100 km blocks, which could very well be located in the observation-barren north.

For example, one particularly "unlucky" seed (555) was associated with only 48 positive cell-weeks in the spatiotemporal test evaluation (compared to the average 127.8 across all seeds). Since AP is calculated as the mean precision@k for the ranks where there are positives, such a low number of positives meant that individual predictions could noticeably sway the AP one way or the other.

Another illustration of the problem is that the relative standard deviation for TF-IDF was a stable 6.1% in the spatial evaluation but 68.0% in the spatiotemporal evaluation. The spatiotemporal results should therefore be considered supplemental.

4 Conclusion

This study set out to investigate whether linguistic features in Swedish place names could improve predictions of chanterelle observations beyond conventional environmental features. My findings support this hypothesis – even when controlling for place name density, which may be linked to where observations are recorded more so than to where chanterelles grow.

More surprising was that the type of linguistic representation matters greatly: A model using sparse character n-gram TF-IDF features beat the control models convincingly across all three evaluation schemes (+20.5-72.5%), while a model with dense, mean-pooled fastText embeddings only barely improved upon the best control model (+0.5-3.9%). This discrepancy may partly reflect information dilution caused by the cell-level aggregation approach employed for fastText embeddings. Since there was a clear north-south divide in the observations data, the success of the TF-IDF model may furthermore stem from its apparent ability to identify not only local environmental features in place names, but also patterns that correlate with latitude.

In the end, the main limitation is that the models predicted observations of chanterelles rather than actual chanterelles. What the linguistic features captured may therefore have been differences in human recording activity as much as nuances of habitat suitability. Future research should primarily focus on bringing predictions closer to the context of the chanterelles themselves, for example by incorporating estimates of observation efforts or more systematic sampling. However, studies could also investigate whether the findings generalize to other species, other datasets, and more fine-grained temporal and geographical resolutions than weeks and 10 x 10 km cells. After all, this study has demonstrated that place names can hold predictive value not just for people interested in predator ranges, oyster beds, and historical forest fires, but also for Swedish mushroom hunters.

References

Sarah Cogos, Lars Östlund, and Samuel Roturier. 2019. Forest fire and indigenous Sami land use: Place names, fire dynamics, and ecosystem change in northern Scandinavia. *Human Ecology*, 47:51–64.

Facebook. 2018. Word vectors for 157 languages.

Maiken Frank-Hansen. 2024. The hunt for the elusive Cantharellales: How to stake the odds in the mycophiles favour, using species distribution modelling and GIS.

Edouard Grave, Piotr Bojanowski, Prakhar Gupta, Armand Joulin, and Tomas Mikolov. 2018. Learning word vectors for 157 languages. In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. European Language Resources Association (ELRA).

Hazelime. 2026. Predicting Chanterelle Observations with Place Names.

Kantarellappen. 2026. Kantarellappen.

A. Kumar, R. Devi, R. Dhalaria, A. Tapwal, R. Verma, S. Rashid, G. M. Elossaily, K. A. Khan, K. T. Chen, and T. Verma. 2024. Nutritional, nutraceutical, and medicinal potential of Cantharellus cibarius fr.: A comprehensive review. *Food Science & Nutrition*, 13(1):e4641.

Lantmäteriet. 2025. Ortnamn Nedladdning, vektor.

Anna McGregor, Carolyn McNamara, C. Ramjan, L. Eisler, and A. Vovides. 2025. Project Eisirean | native oysters: Integrating traditional ecological knowledge in nature restoration work through Scottish place-names. In *ICES Annual Science Conference 2025*. Conference presentation, 15–18 September 2025.

Naturvårdsverket. 2026. Nationella marktäckedata (NMD): Kombinerat basskikt NMD2023 och NMD2018.

Scikit-learn. 2026a. LogisticRegression.

Scikit-learn. 2026b. TfidfVectorizer.

SLU Swedish Species Information Centre. 2026. Artportalen.

SMHI. 2025. Griddad nederbörd- och temperaturdata (PTHBV).

Clara Tattoni. 2019. Nomen omen. toponyms predict recolonization and extinction patterns for large carnivores. *Nature Conservation*, 37:1–16.

A Top TF-IDF n-grams

Table 3 shows the n-grams linked to the most positive and negative TF-IDF coefficients in the temporal evaluation (which covers all grid cells). Note that blank spaces occur in multi-word place names such as *Haga slott* and *Södra Törnskogens naturreservat*.

B Influence of randomization

Figure 3 shows how the choice of randomization seed affected the spatial evaluation (and thus also the spatiotemporal evaluation indirectly, since the seeds were the same). Results are relatively consistent, though Placename_count_log beats Fasttext with seed 333 and both Tfidf and Fasttext with seed 555.

#	Positive	Negative
1	_ku	myren
2	mad	myre
3	rby_	yren
4	maden	yren_
5	made	ávr
6	parke	amoss
7	_kul	yre
8	_kull	amos
9	möl	emo
10	ade	lund_
11	park	_he
12	kul	arp_
13	öll	jáv
14	ulla	myr
15	amn	abo_
16	kulla	jávr
17	ens_	amo
18	ns_	holm_
19	ull	olm_
20	aden	emos

Table 3: N-grams linked to the top 20 TF-IDF coefficients positively and negatively associated with chanterelle observations by order of absolute coefficient size in the temporal evaluation. Blank space has been replaced by underscore for readability.

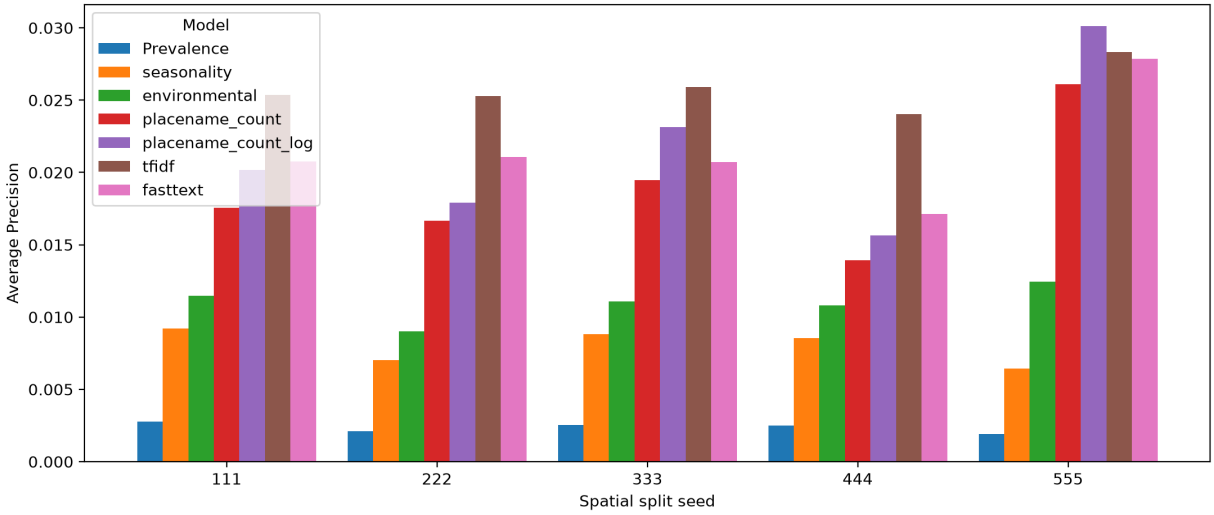


Figure 3: Spatial test AP by random seed for each of the models. Positive prevalence is shown for context.